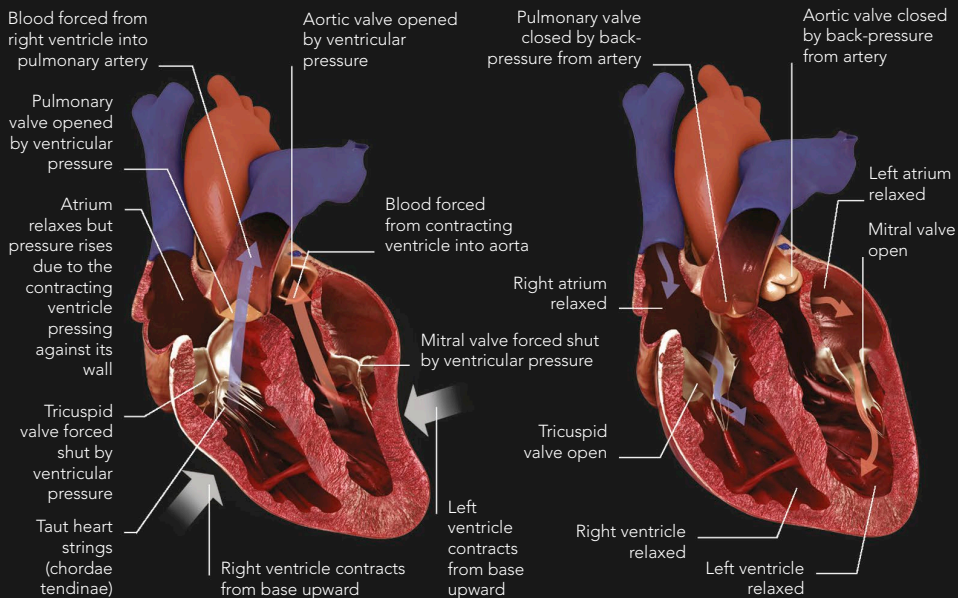
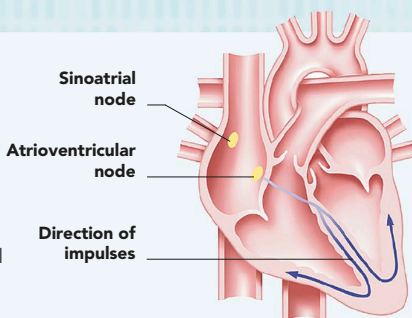


HEARTBEAT SYNCHRONIZATION

Contractions of the atria and the ventricles are synchronized by electrical impulses from the sinoatrial node, the heart's natural pacemaker. Towards the end of diastole, the sinoatrial node sends out electrical impulses. These impulses travel through the atria, making them contract (atrial systole). Some impulses travel to the atrioventricular node, which sends them through conducting fibres to the ventricles, which contract in response (ventricular systole). The electrical impulses then travel back towards the atria. The sinoatrial node then fires again to continue the cycle.



3 CONTRACTION OF THE VENTRICLES (VENTRICULAR SYSTOLE)

During this most active and powerful stage of the heartbeat, the thick cardiac muscle in the ventricle walls contracts, stimulated by electrical impulses relayed by the atrioventricular node. This causes a rise in ventricular pressure, which opens the aortic and pulmonary valves at the exits of the ventricles. Blood is forced out into the main arteries, making the atrioventricular valves snap shut.

4 RELAXATION (EARLY DIASTOLE)

The walls of the ventricles begin to relax, causing ventricular pressure to reduce. The pressure of the recently ejected blood in the main arteries is now high, so both the aortic and pulmonary valves close. This prevents backflow into the ventricles. As ventricular pressure on the atrioventricular valves relaxes, the valves open. This reduces pressure in the atria, allowing blood to enter once again from the main veins.